1. Footprinting Tools conducting Competitive Intelligence

A cybersecurity consultant is assigned to perform a passive security assessment for ABC Technologies the consultant must collect publicly available information without directly interacting with the company's servers. The task is to identify the companys domain registration details, employee email addresses, technology stack and publicly exposed subdomains and explain how each tool helps in competitive intelligence

i) Domain Registration Details

Tool : WHOIS

purpose : * Displays domain owner details

* shows registrar name

* Registration and expiry dates

* DNS / Name servers

* Adminstrative and technical contact information

How it helps :

* identifies technologies uses by the organization

* Helps understand the web infrastructure

ii) Employee Email Addresses

TOOL : Hunter.io, LinkedIn, Google Dorking

purpose : * Finds publicly available employee email addresses

* identifies email naming conventions

* Lists employee profiles

How it helps :

* Helps understand organizational structure

* Assist in identifying public available contact information

iii) Technology stack

TOOL : Builtwith , wappalyzer

purpose : * Detects web server

* programming Language

* CMS and Javascript Libraries

* Analytics tools and Hosting provider

How it helps :

* identifies technologies used by tu organization

* Helps understand tu web infrastructure

4. publicly Exposed subdomains

TOOL : crt.sh, security Trails

purpose : * Finds publicly available subdomains

* uses certificate Transparency logs

* collects DNS information

How it helps :

* Reveals development, testing, mail, vpn and other public services

* supports reconnaissance during security assessments.

2. DNS zone Transfers

During an authorized penetration test a security analyst subsuspects that the organization DNS server allows authorized DNS zone transfers. The task is to verify this using the appropriate command or tool and explain how the retrieved records help identify internal hosts, mail servers and subdomains

DNS zone Transfer

command

dig AXFR example.com @ns1.example.com

Records Retrieved

- A Records
- AAAA Records
- MX Records
- NS Records
- CNAME Records
- TXT Records
- SRV Records
- subdomains

How the Records Help

Internal Hosts

- Reveals server names and IP addresses

Example : server.example.com

db.example.com

Mail servers

- Identifies mail server names
- Helps understand email infrastucture

subdomains Example

- vpn.example.com
- ftp.example.com
- dev.example.com

→ These details assist in reconnanaissance during an authorized penetration test

countermeasures

- Disable public zone transfers
- Allow AXFR only to authorized secondary DNS server
- configure ACLs
- Restrict DNS traffic using firewalls

3. social Engineering Attacks and countermeasures

Employees recieve an email claiming to be from the IT department asking them to verify login credentials through a link because of an urgent security update. The task is to identify the attack and explain suitable countermeasures before responding to the email. Also explain how these actions prevent.

Type of social Engineering Attack

Phishing Attack

The attacker impersonates the IT department and creates urgency to trick employees into revealing usernames and passwords

countermeasures

verify the sender

- check the sender's email address
- Ensure it belongs to the offical company details

DO Not click suspicious Links

- Hover over click links to inspect tu destination
- visit tu official website manually if necessary

Never share credentials

- Legitimate IT department never request passwords through email

contact the IT Department

- verify the request through official communication channels

Report the Email

- Inform the IT or cybersecurity team immediately

Enable multi-factor Authentication (MFA)

- provides an additional security layer even if passwords are compromised

Security Awareness Training

- Learn to recognize phishing emails and suspicious requests

How these Actions prevent credential that

- Verify the sender detects fake emails
- Avoiding suspicious links prevents visiting fake login pages
- not sharing passwords protects accounts
- Reporting phishing helps protect other employees
- security awareness reduces the success of phishing attacks.